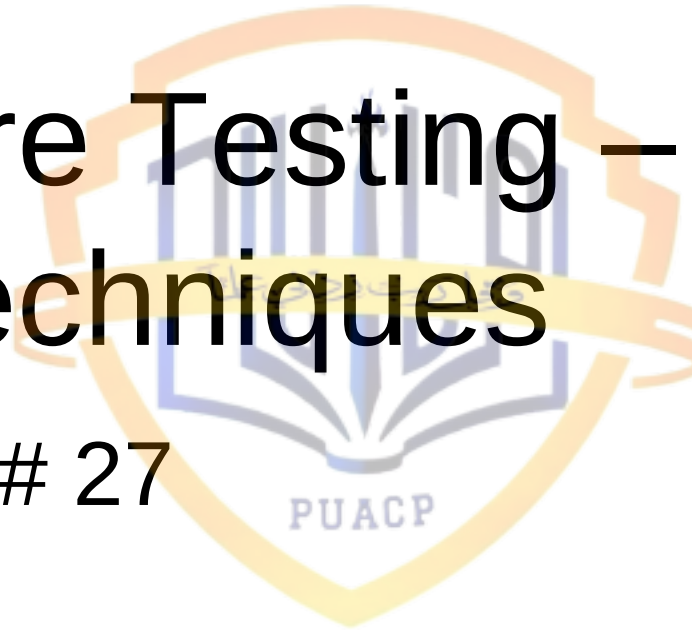




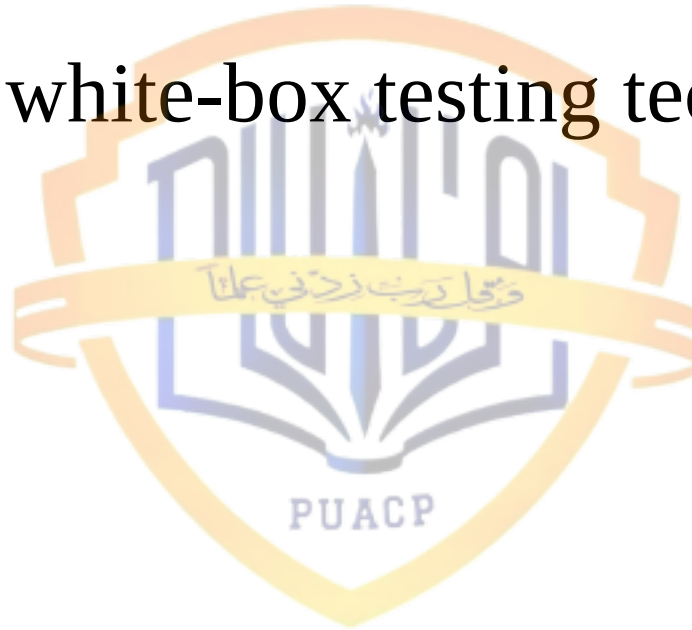
Software Testing – 5 – BBT Techniques

Lecture # 27



Recap

- Discussed white-box testing techniques in detail



Today's Lecture

- We'll have some more discussion on black-box testing
- Specifically, we'll talk about different techniques, which are classified as black-box techniques

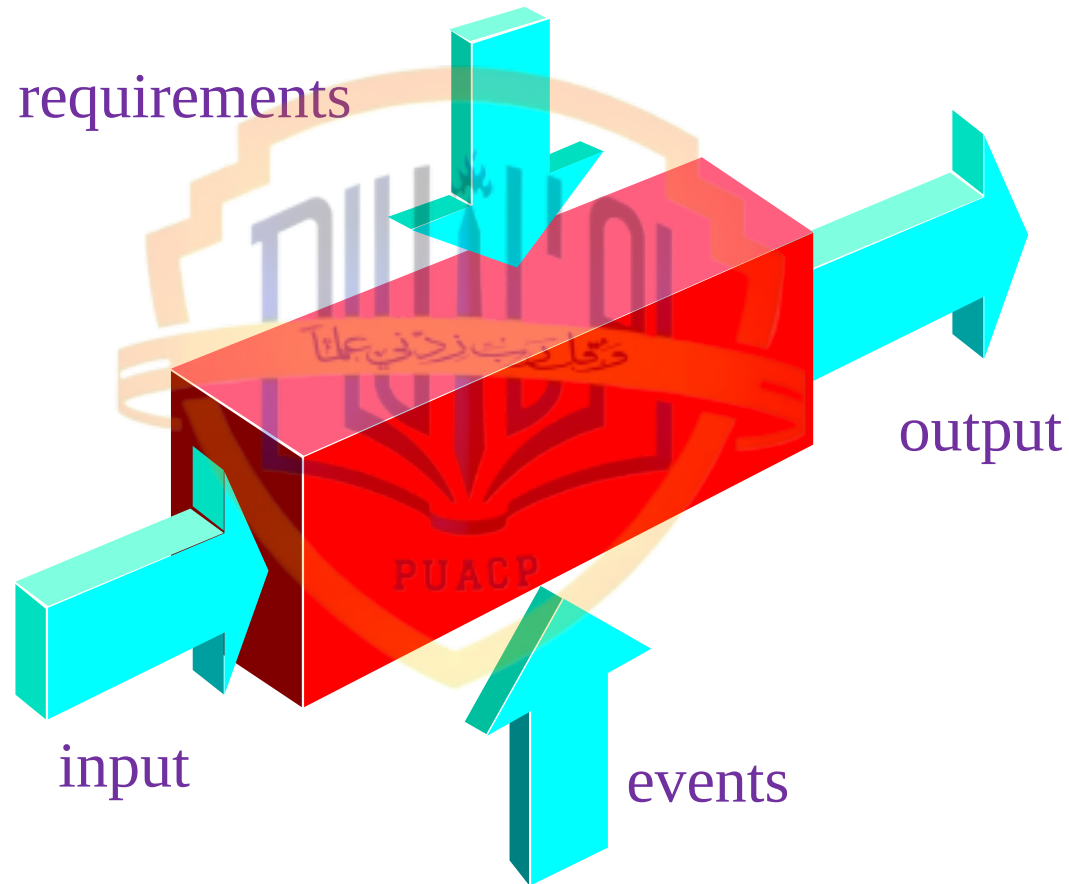
Black-box Testing



Black-box Testing - 1

- BBT focuses on the functional requirements of the software
- It enables the software engineer to derive sets of input conditions that will fully exercise all functional requirements for a program
- It is not an alternative to WBT, rather a it is a complementary approach that is likely to uncover a different class of errors

Black-Box Testing

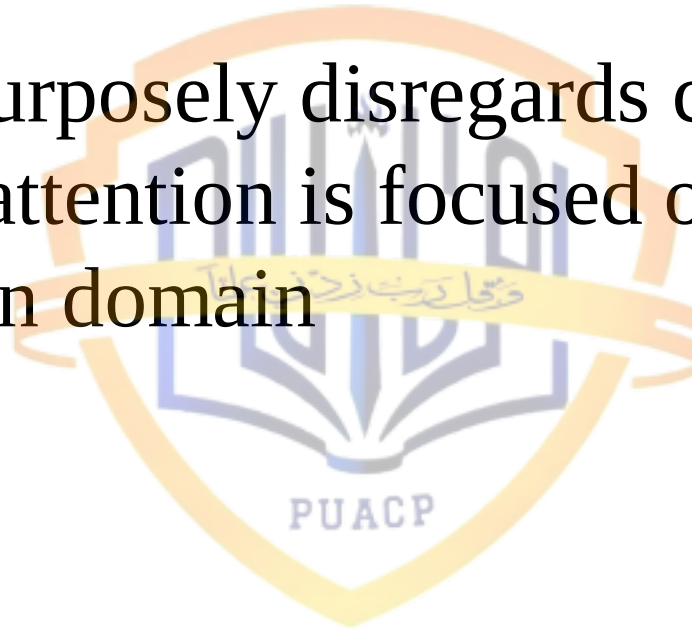


Black-box Testing - 2

- BBT attempts to find errors in the following categories
 - Incorrect or missing functions
 - Interface errors
 - Errors in data structures or external data base access
 - Behavior or performance errors
 - Initialization and termination errors

Black-box Testing - 3

- As BBT purposely disregards control structure, attention is focused on the information domain



Black-box Testing - 4

- Tests are designed to answer the following questions
 - How is functional validity tested?
 - How is system behavior and performance tested?
 - What classes of input will make good test cases?

Black-box Testing - 5

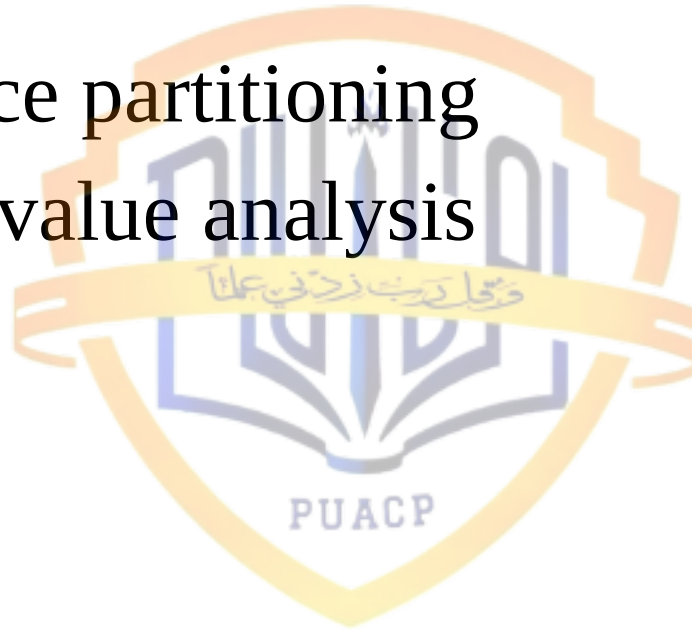
- Is the system particularly sensitive to certain input values?
- How are the boundaries of a data class isolated?
- What data rates and data volume can the system tolerate?
- What effect will specific combinations of data have on system operation?

Black-box Testing - 5

- By applying BBT, we derive a set of test cases that satisfy the following criteria
 - Test cases that reduce the number of additional test cases that must be designed to achieve reasonable testing
 - Test cases that tell us something about the presence or absence of classes of errors, rather than an error associated only with the specific test at hand

BBT Techniques

- Equivalence partitioning
- Boundary value analysis



Equivalence Partitioning



Equivalence Partitioning - 1

- Equivalence partitioning is a black-box testing method that divides the input domain of a program into classes of data from which test cases can be derived
- Equivalence partitioning strives to define a test case that uncovers classes of errors, thereby reducing the total number of test cases that must be developed

Equivalence Partitioning - 2

- An equivalence class represents a set of valid or invalid states for input conditions
- Typically, an input condition is
 - A specific numeric value
 - A range of values
 - A set of related values
 - A boolean condition

Guidelines for Equivalence Classes - 1

- If an input condition specifies a range, one valid and two invalid equivalence classes are defined
- If an input condition requires a specific value, one valid and two invalid equivalence classes are defined

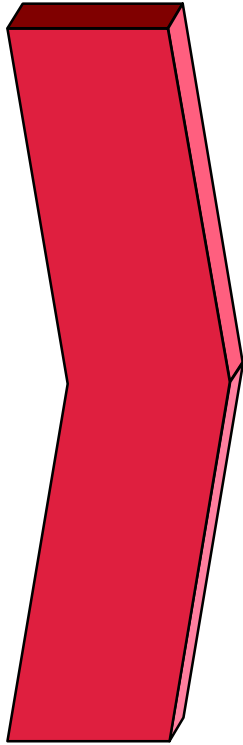
Guidelines for Equivalence Classes - 2

- If an input condition specifies a member of a set, one valid and one invalid equivalence class are defined
- If an input condition is boolean, one valid and one invalid class are defined

Equivalence Partitioning



Sample Equivalence Classes



Valid

data

user supplied commands

responses to system prompts

file names

computational data

physical parameters

bounding values

initiation values

output data formatting

responses to error messages

graphical data (e.g., mouse picks)

Invalid

data

data outside bounds of the program

physically impossible data

proper value supplied in wrong place

Discussion on Equivalence Partitioning



Boundary Value Analysis



Boundary Value Analysis - 1

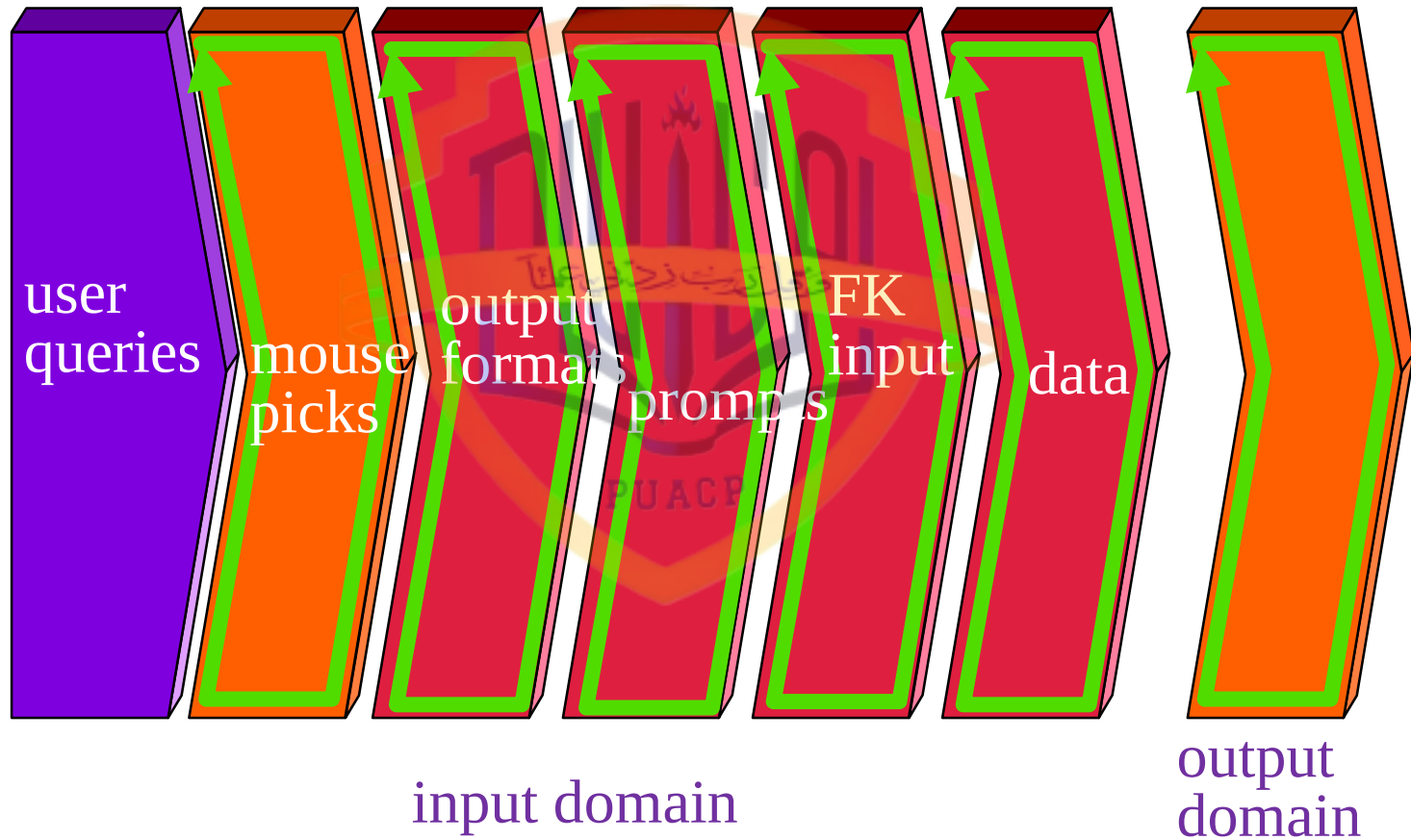
- Boundary value analysis (BVA) is a testing technique, which leads to a selection of test cases that exercise bounding values
- This is because that for reasons not clearly completely clear, a greater number of errors tends to occur at the boundaries of the input domain rather than in the “center”

Boundary Value Analysis - 2

- BVA complements equivalence partitioning testing technique



Boundary Value Analysis



State Testing

- Uses equivalence partitioning techniques

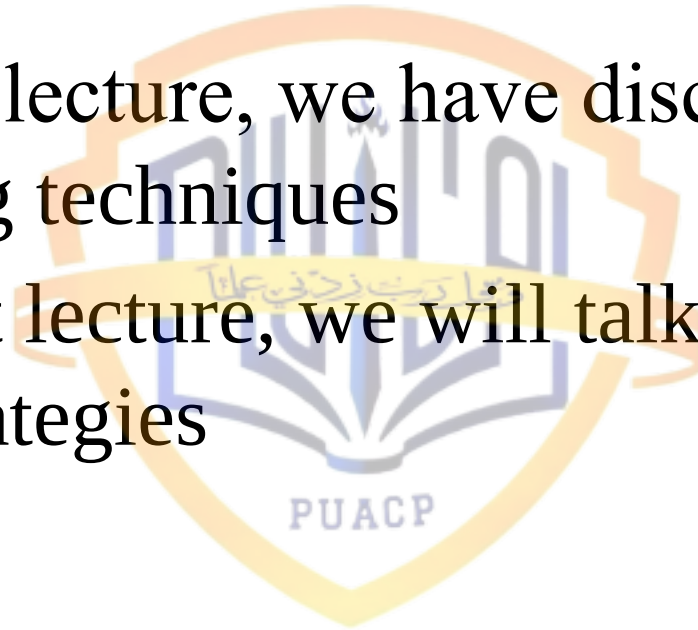


Differences between BBT and WBT



Summary

- In today's lecture, we have discussed black-box testing techniques
- In the next lecture, we will talk about testing strategies



References

- Software Engineering: A Practitioner's Approach by Roger S. Pressman (Chapter 17.3-17.8)

